



TECHMINDS ACADEMY

CSS Mastery

Web Styling Curriculum · Beginner to Confident

A structured 12-week programme for students who have completed the TechMinds HTML curriculum. From your first CSS rule to building a fully responsive, animated multi-page website.

12 Weeks	5 Stages	55+ Lessons	8 Projects	1 Prerequisite
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Selectors & Specificity · Box Model · Typography & Colour · Flexbox · CSS Grid · Responsive Design · Transitions & Animations

LEARNING PATHWAY

Programme Overview

Stage	Focus	Duration	Level	Prerequisite
Stage 1	CSS Foundations — selectors, syntax	2 weeks	Beginner	HTML course
Stage 2	Box Model & Spacing	2 weeks	Beginner	Stage 1
Stage 3	Typography & Colour	2 weeks	Intermediate	Stage 2
Stage 4	Layout — Flexbox & CSS Grid	3 weeks	Intermediate	Stage 3
Stage 5	Responsive Design, Animations & Capstone	3 weeks	Advanced	Stage 4

CURRICULUM RATIONALE

Key Decisions Explained

Why CSS after HTML?	CSS has no meaning without HTML to style. Students who attempt CSS without solid HTML foundations write guesswork code they cannot debug. The TechMinds track is sequential by design — each course builds directly on the last.
Why dedicate 3 weeks to layout?	Flexbox and CSS Grid are the skills employers test most in junior developer interviews. Three weeks ensures the student builds genuine confidence — not just surface familiarity — with both layout systems.
Why teach mobile-first?	Over 60% of web traffic worldwide is mobile. Writing CSS mobile-first is the industry standard. Students who learn desktop-first must unlearn bad habits later. This curriculum builds the right instinct from the start.
What comes after this course?	CSS Mastery feeds directly into JavaScript Fundamentals — the next course in the TechMinds web development track. Students finishing this programme are ready to add interactivity to the pages they can now build and style.

CSS Foundations

How CSS works and how to connect it to HTML

This stage assumes the student has completed the HTML curriculum. CSS is introduced from the ground up — what it is, how the browser reads it, and how to write rules that change the appearance of HTML elements. By the end of Week 2 the student will be styling real pages with confidence.

Week 1

Introduction to CSS

5 lessons · 5 hrs

- 01 What is CSS and what does it do?
- 02 Three ways to add CSS: inline, internal, external
- 03 Why external stylesheets are the correct approach
- 04 Linking a CSS file to an HTML page with link
- 05 CSS syntax: selectors, properties, and values
- 06 Your first CSS rules — changing colour, font, and size

PROJECT

Style the Week 1 HTML page from the HTML course using an external stylesheet

Week 2

Selectors & Specificity

5 lessons · 6 hrs

- 01 Element selectors — targeting tags directly
- 02 Class selectors — reusable styles with .classname
- 03 ID selectors — unique styles with #idname
- 04 Grouping selectors — styling multiple elements at once
- 05 Descendant and child selectors
- 06 Specificity — why some rules win over others

PROJECT

A styled article page using element, class, and ID selectors correctly

The Box Model & Spacing

Understanding how every element takes up space on the page

The box model is the single most important concept in CSS. Every element on a page is a box — with content, padding, border, and margin. Students who truly understand the box model can control layout with precision. This stage also introduces the browser developer tools for inspecting live pages.

Week 3

The Box Model

5 lessons · 6 hrs

- 01 Content, padding, border, and margin — the four layers
- 02 How width and height interact with padding and border
- 03 box-sizing: border-box — the modern standard
- 04 Inspecting the box model in Chrome DevTools
- 05 Shorthand values: margin: 10px 20px vs margin: 10px 20px 10px 20px
- 06 Collapsing margins — the behaviour every student gets wrong

PROJECT

A card component — title, description, image, and button with precise box model spacing

Week 4

Display, Visibility & Overflow

5 lessons · 5 hrs

- 01 Block vs inline vs inline-block — how elements behave
- 02 Changing display type with the display property
- 03 visibility: hidden vs display: none — the difference
- 04 overflow: hidden, scroll, auto — controlling content overflow
- 05 The border property — style, width, color, and radius
- 06 border-radius — rounded corners and circle shapes

PROJECT

A profile card with a circular avatar, rounded corners, and controlled overflow

Stage 03 · Intermediate

2 Weeks · Weeks 5 – 6

Typography & Colour

Making text beautiful and using colour with intention

Typography and colour are where design meets code. This stage teaches the student how to choose and apply fonts using Google Fonts, control every aspect of text appearance, and work with the full range of CSS colour formats — hex, RGB, HSL, and opacity.

Week 5

Typography

5 lessons · 5 hrs

01 font-family — web safe fonts and font stacks

02 Importing and using Google Fonts

03 font-size: px, em, rem — when to use each

04 font-weight, font-style, and font-variant

05 line-height, letter-spacing, and word-spacing

06 text-align, text-decoration, text-transform, and text-shadow

PROJECT

A typographically styled blog post using Google Fonts, rem sizes, and line-height

Week 6

Colour & Backgrounds

5 lessons · 5 hrs

01 CSS colour formats: named, hex, rgb(), hsl()

02 Opacity with rgba() and the opacity property

03 background-color and background-image

04 background-size: cover and contain

05 background-position and background-repeat

06 CSS gradients: linear-gradient and radial-gradient

PROJECT

A hero section with a full-width background image, gradient overlay, and styled text

Stage 04 · Intermediate

3 Weeks · Weeks 7 – 9

Layout — Flexbox & Grid

Building modern, responsive page layouts

Layout is where CSS becomes truly powerful — and where most beginners get stuck. This stage dedicates three full weeks to Flexbox and CSS Grid, the two modern layout systems that every web developer must know. By the end, students will build multi-column layouts with confidence.

Week 7

Flexbox — Part 1

5 lessons · 6 hrs

01 What is Flexbox and why it replaced float layouts

02 display: flex — activating the flex container

03 flex-direction: row and column

04 justify-content — aligning items along the main axis

05 align-items — aligning items along the cross axis

06 gap — spacing between flex items

PROJECT

A horizontal navigation bar and a vertically centred hero section using Flexbox

Week 8

Flexbox — Part 2

5 lessons · 6 hrs

01 flex-wrap — allowing items to wrap onto new lines

02 flex-grow, flex-shrink, and flex-basis

03 The flex shorthand property

04 align-self — overriding alignment for a single item

05 order — reordering items without changing the HTML

06 Building a real card grid layout with Flexbox

PROJECT

A 3-column responsive card grid — services, portfolio, or team members

Week 9

CSS Grid

5 lessons · 7 hrs

01 display: grid — activating the grid container

02 grid-template-columns and grid-template-rows

03 The fr unit — fractional space in a grid

04 grid-gap and the gap shorthand

05 grid-column and grid-row — spanning across cells

06 Flexbox vs Grid — when to use which

PROJECT

A full page layout — header, sidebar, main content, and footer using CSS Grid

Stage 05 · Advanced

3 Weeks · Weeks 10 – 12

Responsive Design, Transitions & Capstone

Making websites work on every screen size

Over half of all web traffic comes from mobile devices. This final stage teaches responsive design with media queries, smooth hover transitions, and CSS animations. The course closes with a capstone project — a fully responsive, styled multi-page website built entirely with HTML and CSS.

Week 10

Responsive Design & Media Queries

5 lessons · 6 hrs

01 What is responsive design? Mobile-first vs desktop-first

02 The viewport meta tag and why it is required

03 Media queries: @media and breakpoints

04 Common breakpoints: 480px, 768px, 1024px, 1280px

05 Responsive typography with clamp()

06 Responsive images: max-width: 100% and object-fit

PROJECT

Make the Week 9 page layout fully responsive — stacks on mobile, side-by-side on desktop

Week 11

Transitions, Hover Effects & Animations

5 lessons · 6 hrs

01 CSS transitions — smooth property changes on hover

02 transition-property, duration, timing-function, delay

03 Hover states — buttons, cards, and navigation links

04 CSS transforms: scale, rotate, translate, skew

05 CSS animations with @keyframes

06 animation-name, duration, iteration-count, and direction

PROJECT

An interactive card with hover lift, button colour transition, and a loading spinner animation

Week 12

Capstone Project

6 sessions · 10 hrs

01 Planning the design — colours, fonts, spacing, layout

02 Building the Home page — hero, features section, footer

03 Building the About page — team grid, image and text layout

04 Building the Services or Portfolio page — card grid layout

05 Building the Contact page — styled form and map embed

06 Responsive audit — test on mobile, tablet, and desktop

PROJECT

A fully responsive 4-page website styled entirely with CSS — submitted and reviewed

ASSESSMENT & COMPLETION

How Students Are Evaluated

Component	Description	Weight
Weekly Projects (×8)	Styled page or component built from scratch each week	40%
Stage Assessments (×5)	Short quiz and CSS code review at end of each stage	30%
Capstone Website	Fully responsive 4-page website — styled, animated, reviewed	30%

LEARNING OUTCOMES

What Students Can Do on Completion

On completing this programme, the student will be able to:

- Write clean, organised CSS using external stylesheets and proper selectors
- Explain and apply the CSS box model to control spacing and layout precisely
- Style typography using Google Fonts, rem units, and line-height
- Work with all CSS colour formats including hex, rgb(), hsl(), and opacity
- Build any layout using Flexbox and CSS Grid confidently
- Write media queries and build fully responsive pages from mobile up
- Apply CSS transitions and animations to create smooth, professional interactions
- Build and submit a complete, responsive 4-page styled website

TOOLS & RESOURCES

Everything the Student Needs

Tool	Purpose	Cost
VS Code	Primary code editor for writing CSS	Free
Live Server	Auto-refreshes browser on every file save	Free
Chrome DevTools	Inspect box model, computed styles, and responsive views	Built-in
Google Fonts	Free web font library used throughout the typography module	Free
CSS Validator (W3C)	Check CSS for errors and standards compliance	Free
GitHub Pages	Host and share the capstone website publicly for free	Free

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